Math 54 Discussion Worksheet

Name: ____________________ ID: ____________

Problem 1

Express the following higher-order differential equations as matrix systems in normal form:

1. $(1-t^2)y'' - 2ty' + 2y = 0$
2. $y'' - ty = 0$
3. $y'' + \frac{1}{t}y' + \left(1 - \frac{n^2}{t^2}\right)y = 0$

Problem 2

Prove that the operator given by $L[x] = x' - Ax$ is a linear operator. Here x is an $n \times 1$ column vector and A is an $n \times n$ matrix.

Problem 3
Let $X(t)$ be the fundamental matrix for the system $x' = Ax$. Show that $x(t) = X(t)X^{-1}(t_0)x_0$ is the solution to the intial value problem $x' = Ax, x(t_0) = x_0$.